

Beyond Relevance: On the Relationship Between Retrieval and RAG Information Coverage

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Abstract

Retrieval-augmented generation (RAG) systems combine document retrieval with a generative model to address complex information-seeking tasks like report generation. While the relationship between retrieval quality and generation effectiveness seems intuitive, it has not been systematically studied. We investigate whether upstream retrieval metrics can serve as reliable early indicators of the final generated response’s information coverage. Through experiments across two text RAG benchmarks (TREC NeuCLIR 2024 and TREC RAG 2024) and one multimodal benchmark (WikiVideo), we analyze 15 text retrieval stacks and 10 multimodal retrieval stacks across four RAG pipelines and multiple evaluation frameworks (Auto-ARGUE and MiRAGE). Our findings demonstrate strong correlations between coverage-based retrieval metrics and nugget coverage in generated responses at both topic and system levels. This relationship holds most strongly when retrieval objectives align with generation goals, though more complex iterative RAG pipelines can partially decouple generation quality from retrieval effectiveness. These findings provide empirical support for using retrieval metrics as proxies for RAG performance.

CCS Concepts

• **Information systems** → **Information retrieval diversity**; *Summarization*; • **Computing methodologies** → Natural language generation.

Keywords

Retrieval-augmented Generation, Search Result Diversification, Generation, Retrieval, Correlation, Information Coverage

1 Introduction

As information systems have moved from merely providing source information for human consumption to providing a synthesized text that summarizes the source information, system architecture has changed dramatically. When the system outputs a ranked list or a search engine result page (SERP), gathering documents that are most likely to be relevant to the user query has been the typical design goal of the systems. The community generally refer to this class of problems as *ad hoc retrieval*.

However, when users expect only a piece of text synthesized from the source information, a succinct and well-organized summary is usually more preferable than a simple concatenation of relevant documents [67]. Similar and related information are expected to be grouped, rephrased, or merged [6]; multiple facets should be included if they are all relevant to the user query [42]. Such preference implies that source documents providing the same set of relevant information, while still equally relevant to the query, are not as useful after the first one has been processed by the downstream summarization model, usually powered by a large language model (LLM) [3, 11, 17, 18, 64]. Therefore, the objective of the information system has changed – it should provide a summary of information covering multiple aspects or facets without redundancy. This kind of retrieval problem is generally being referred as *report generation* [34].

These systems are usually implemented with retrieval-augmented generation (RAG), which combines the strengths of document retrieval with LLM generation to address complex information seeking tasks. Ad hoc retrieval, in this case, becomes an upstream component in the system by providing source information to generation model. However, given the final objective of report generation, it *should* be sufficient for the upstream retrieval model to provide a relatively small number of documents that cover all needed aspect in the final generation. To date, these LLMs are not able to faithfully leverage all advertised context window, exhibiting information loss [35], [16] and hallucination [27] when exceeding certain effective limit. In fact, as long as the LLM has an effective context window, there is need to filter the collection to a smaller subset of documents for the downstream model. Therefore, minimizing the number of document the generation model needs to process *should* provide benefits to the final generated responses. The problem of retrieving for multiple aspects with penalization on redundant information has been studied in prior retrieval literature on diversity ranking [8], which is equivalent to information coverage [68].

However, the relationship between information coverage of the upstream retrieval and the final generation quality, though logical, has not been systematically studied. Furthermore, end-to-end evaluation of generated responses requires running complete RAG pipelines, which incurs substantial computational cost. Unlike document-level judgments, judgments of generated responses

are not easily reusable, so evaluating them incurs substantial further costs in the form of collecting new judgments from a human or LLM. Moreover, the LLM itself adds variability and noise, as different LLMs and generation strategies can produce divergent outputs even when given identical retrieved context, resulting in noisy signals when attributing effectiveness to each component in the pipeline [4, 7].

In this paper, we aim to provide a systematic study on the relationship between upstream retrieval and the downstream generation quality. Particularly, we focus on information coverage (usually materialized as nugget coverage in recent literature [1, 47]) of the final generation quality since it is the primary purpose of employing a retrieval model in the generation pipeline. Other qualities, such as fluency and faithfulness, can already be attributed to the LLM itself [41], thus are excluded in this study. Particularly, we aim to answer the question: “Is the upstream retrieval quality an early indicator of the information coverage of the downstream generation responses?” With our empirical evidence, we show that there is a strong relationship between the two stages. This evidence provides empirical grounds for simplifying the evaluation of information coverage to focus on the upstream retrieval model, reducing both computational costs and experimental noise. Such a simplification is common in the literature such as focusing on recall for first-stage retrieval model [25] and precision at one for question-answering and conversational search (which need to be announced to users through audio).

Our analysis operates at multiple levels. Topic-level analysis examines whether better retrieval results (or a ranked list) for a specific query lead to a better generated response. System-level analysis assesses whether employing a more effective retrieval models yields a more effective RAG pipeline in general, measuring by information coverage. For robustness, we examine these relationships across different generation pipelines, from simple retrieve-then-generate approaches [15, 31, 64] to more complex multi-query and iterative strategies [2, 9], to understand whether generation complexity can compensate for weaker retrieval. Our contributions are:

- We demonstrate that nugget-oriented retrieval metrics serve as reliable indicators of RAG information coverage, with strong correlations observed at both topic and system levels across text and multimodal benchmarks.
- We show that RAG pipeline complexity affects the retrieval-generation relationship. Simpler linear pipelines benefit directly from retrieval improvements, while complex iterative pipelines can partially decouple generation quality from retrieval effectiveness by adapting queries to retrieval system capabilities.
- We validate our findings across multiple generation strategies (GPT-Researcher, Bullet List, LangGraph), evaluation frameworks (Auto-ARGUE, MiRAGE), and modalities (text and video), demonstrating the robustness and generalizability of coverage-based retrieval metrics as proxies for RAG performance.

2 Background

2.1 Retrieval and Evaluation

Traditional adhoc retrieval evaluation has focused on document relevance. Metrics such as MRR, MAP, and nDCG measure the quality of ranking based on a model’s ability to put relevant document at the top of the ranking with different browsing models [26, 50].

However, retrieval models embedded in a RAG pipeline for complex information seeking tasks need to retrieve a broad coverage of information to satisfy the complexity of the information need. Therefore, coverage metrics, such as Sub-topic Recall [12] and α -nDCG [13] become more appropriate, as they evaluate how well a retrieval system gathers information that collectively addresses all aspects of the user’s information need [10]. α -nDCG [13] incorporates both relevance and diversity, penalizing redundant information while rewarding novel relevant content. Sub-topic recall explicitly measures whether retrieval results cover multiple facets of the information needed [12]. These coverage metrics align with the goals of RAG systems, since generation requires comprehensive information to produce complete and accurate responses.

These metrics were developed for retrieval diversification tasks, such as TREC Interactive Track [45] and TREC Novelty Track [54]. These tasks generally aim to provide a diverse set of retrieval results covering different potential user intent given a short, ambiguous query. TREC Complex Answer Retrieval Track [43] needs diversity for answering a simple question with a complex answer, covering multiple facet of the question. While methods from these efforts are certainly applicable in report generation task, the goal is improving *coverage* of information that the user would like to receive instead of solely covering results *as broad as possible*. Therefore, while metrics from diversification literature are useful, we view diversification as one of the approaches to improve information coverage.

2.2 RAG Pipelines

To synthesize a generated response for satisfying complex information needs, RAG systems usually consist of two primary components: an upstream retrieval component that identifies and ranks relevant documents from a corpus and a downstream generation component that synthesizes information from retrieved documents to produce coherent responses. While this basic architecture is consistent across RAG systems, the strategies for combining retrieval and generation vary significantly in complexity [23].

2.2.1 Linear RAG Pipelines. The simplest RAG architecture follows a retrieve-then-generate pattern. This is where the system retrieves documents once based on the user’s query and then generates a response from the retrieved context. Crucible [15] and GINGER [31] are two examples, where relevant information is extracted from the top- k from a diverse ranking. After retrieval, such systems follow a multi-step response generation stage for the retrieved content via nugget extraction and detection, clustering, ranking, summarization, and fluency enhancement.

A natural extension introduces sub-query generation, where the system decomposes complex queries into multiple sub-queries to retrieve more diverse information. GPT Researcher (GPT-R) [18, 19] and Bullet List (extractive approach in Yang et al. [64]) generate multiple queries to gather information from different perspectives

before generation. While sub-query generation could be seen as part of the retrieval stack, similar to query expansion [33] or query rewriting [9] techniques and pseudo relevance feedback in traditional retrieval pipeline, in this paper we treat it as part of the generation pipeline, as it typically involves LLM decomposition that is tightly coupled with the generation process.

2.2.2 Iterative RAG Pipelines. More sophisticated RAG systems employ iterative strategies that alternate between retrieval and generation multiple times, such as Self-RAG, Im-rag, SIM-RAG, etc[3, 20, 24, 29, 61, 62]. After an initial retrieval step, the system analyzes retrieved documents to identify information gaps, then performs additional retrieval iterations to fill those gaps. This iterative process continues until the system determines it has sufficient information to generate a comprehensive response. This iterative retrieval could also be viewed as part of the retrieval stack, similar to relevance feedback mechanisms in classical IR [33]. But the integration with generation logic and the LLM decision making about when and what to retrieve each iteration leads us to consider it part of the generation pipeline for our analysis.

2.3 RAG Evaluation

RAG evaluation presents unique challenges because it assesses generated responses. In the context of tasks like report generation where document citations are required, the evaluation must assess both whether the generated response accurately addresses the user’s information needs and whether the generated response reflects and cites information in the retrieved documents. Various evaluation frameworks have emerged to address these challenges, such as ARGUE [40], AutoNuggetizer [47], EXAM [21], MiRAGE [39], and RUBRIC [22].

Nugget-based evaluation provides an framework to measuring information coverage. Human assessors identify atomic units of information (nuggets) that constitute a complete answer, then evaluate generated responses based on how many nuggets they contain.

Auto-ARGUE [57] applies ARGUE [40] nugget evaluation to RAG systems by representing nuggets as question-answer pairs. Given a query and a set of retrieved documents, assessors create QA pairs that capture distinct pieces of relevant information. Generated responses are then evaluated by checking which QA pairs they answer correctly. This approach provides fine-grained coverage measurement without decomposing the generation into subclaims.

MiRAGE [39] takes a different approach by representing nuggets needs as claims that can be decomposed into subclaims. The framework evaluates both whether subclaims are attested in the generated response (recall) and whether attested claims are properly cited with correct sources. MiRAGE’s citation recall metric aligns with our goals since a nugget should both appear in the generated response and be grounded in the correct source document.

We use the Auto-ARGUE [57] and MiRAGE [39] to investigate whether retrieval systems that gather more comprehensive information lead to generated responses with higher information coverage. We consider the relationship between information coverage of retrieved documents and of generated responses at both the topic level (i.e., whether improving retrieval coverage on a specific topic leads to better generation coverage on that topic) and at the system level (i.e., whether retrieval systems that have better coverage on

average lead to generated responses that have better coverage on average).

3 Research Questions

To investigate the relationship between retrieval and RAG information coverage, we formulate the following research questions to guide the exploration.

RQ1 For a given RAG pipeline, does an input rank list that covers more information lead to a more effective generated response for a specific topic?

To address this question, we investigate the correlation between retrieval effectiveness and the generated response at the topic level (i.e., correlation on the topic scores instead of first averaging over the topics) over various retrieval stacks. This analysis provides insight into how retrieval effectiveness on a specific topic directly impacts the downstream generated response’s information coverage.

RQ2 For a given RAG pipeline, does using a more effective retrieval system lead to a more effective RAG system on average across topics?

Practically, when building a RAG pipeline, practitioners need to pick a retrieval system as the pipeline component. This system should be effective for all (or on average) user queries. Therefore, we investigate the correlation between a retrieval system’s average evaluation metric values and the generation response’s average nugget coverage. We investigate this relationship both within and across domains, as well as on the same and distinct retrieval task objectives. Such experiments allow us to simulate a practical system design process where the retrieval system’s effectiveness cannot be measured on the topics that users would issue to the RAG system.

RQ3 Can a more complex RAG pipeline compensate for a less effective retrieval system, decoupling generation quality from retrieval effectiveness?

RAG pipelines vary in complexity, ranging from receiving a fixed ranked list of documents from the retrieval model based on the user’s query to iteratively reflecting and generating queries to issue to the retrieval model. We investigate whether the relationships found in RQ1 and RQ2 are still applicable to more complex RAG pipelines.

RQ4 Do these relationships hold across different RAG evaluators?

Given that generation tasks like report generation typically do not have a single correct answer, RAG evaluation requires automatic evaluation systems to score the generated responses. We further investigate the robustness of our findings on the previous research questions across evaluation approaches.

RQ5 Do these relationships hold in multimodal RAG?

Finally, we investigate our findings on a multimodal report generation task for further evidence on the generalizability of our findings. Since multimodal and video RAG experiments are much more expensive to conduct, we consider a single, representative generation system in our experiments.

4 Experiment Setup

To systematically investigate the five research questions, we conduct our experiments across two text and one multimodal generation task to support both within-dataset and cross-dataset analysis.

4.1 Evaluation Datasets

For end-to-end RAG evaluation, we use TREC NeuCLIR 2024 Report Generation Pilot Task (NeuCLIR24) [34] and TREC RAG 2024 (RAG24) [47, 55, 56] as our two text RAG tasks because of their nugget annotations, which enable information coverage evaluation.

NeuCLIR24 is a multilingual report generation task where the user input is a problem statement that expresses a rich information need in a paragraph of text, along with a background that describes the profile of the user. NeuCLIR24 has 19 topics assessed for the report generation pilot. The document collection contains more than 10 million news articles extracted from Common Crawl in Chinese, Persian, and Russian. RAG24 is a question-answering task that requires the system to retrieve multiple documents as supporting evidence for the final answer. RAG24 uses MS MARCO Segment v2.1 [46] as the document collection with 55 judged queries. These questions, for example, “*why are trade-offs so important to the success of a business?*”, while not as simple as factual questions, can often be answered by a single document. However, there are multiple documents that contain acceptable answers that can be retrieved.

Additionally, we use WikiVideo [38], an event-centric article writing task using video as the supporting documents, to further verify our findings with multimodal RAG. WikiVideo contains 109K videos and 57 topics with an average of 8 relevant videos per topic (maximum 10 videos)

We refer to the user input for the RAG tasks all as *queries*, and acknowledge the nuanced differences between the kind of user input in each task. To evaluate the retrieval models, we use the nuggets in each collection to form a nugget-based qrels that treats each nugget in a topic as an aspect or group. This format is supported by various retrieval evaluation toolkits, including trec-eval, ndeval, and ir-measures [37]. Additionally, we also include MultiVent 2.0 [30], containing the superset of WikiVideo videos, as additional retrieval tasks for analysis.

4.2 Retrieval Systems

We include retrieval systems that combine different first-stage retrieval and reranking methods. For consistency, when reranking, we always rerank the top 100 documents from the first-stage model. These retrieval stacks are served using RoutIR [65], a serving package that wraps retrieval models with pipeline APIs to support interactions between retrieval and generation systems.

For NeuCLIR24 and RAG24, we use the following first-stage retrieval models, covering common first-stage retrieval architectures:

- BM25 [51] as our lexical retrieval baseline.
- PLAID-X [63], a strong multilingual late-interaction model that achieves the best first-stage effectiveness in TREC NeuCLIR 2023 and 2024.
- LSR. We use MILCO [44], a multilingual learned sparse retrieval (LSR) model that projects text into a shared English lexical space for NeuCLIR24 and SPLADEv3 [32], a strong

English LSR model, for RAG24.

- Qwen3-8B Embed [60, 66], a multilingual dense embedding model built on the Qwen3.
- 3-way RRF, which combines results from PLAID-X, LSR, and Qwen3-8B Embed with Reciprocal Rank Fusion [14]

To further improve retrieval effectiveness, we employ the two rerankers to rerank the first-stage retrieval results from each method:

- Qwen3-8B Reranker [66], a multilingual pointwise reranking model built on Qwen3 [60], with state-of-the-art effectiveness in various retrieval tasks.
- Rank1-7B [58], a pointwise reasoning reranking model that allows test-time scaling based on Qwen2.5.

For WikiVideo, we use the following multimodal retrieval models as the first-stage retrieval model that leverage different modalities:

- CLIP [48] is a pretrained text-image model. To retrieve videos, we take 16 frames and use the maximum similarity between the query-frame pairs.
- LanguageBind [69], a multimodal framework that uses language to embed different modalities in a shared space.
- MMMORRF [52], a video retrieval system that combines visual, audio, and text features with modality-aware weighted reciprocal rank fusion.
- Video-ColBERT [49], a late-interaction text-to-video model.
- OmniEmbed [36], a omnimodal dense encoder built on Qwen2.5 Omni [59].

For WikiVideo, we employ RankVideo [53], a reasoning reranker for pointwise reranking in text-to-video retrieval.

Overall, we investigate 15 retrieval stacks (5 first-stage models with and without the 2 rerankers) for the NeuCLIR24 and RAG24 tasks; for WikiVideo, we use 10 stacks.

4.3 RAG Pipelines

We use four RAG pipelines for NeuCLIR24 and RAG24 tasks: GPT-Researcher (GPT-R) [18, 19] with one and three queries, Bullet List [64], and LangGraph [17]. All generation pipelines are given a processing budget of 50 documents to enable a fair comparison between pipelines. All text RAG pipelines use Llama-70B-Instruct as the backbone LLM.

GPT-Researcher (GPT-R) [18, 19] is used as a cascade system that retrieves a seed set of documents given the query, generates additional sub-queries, retrieves documents with the original query and sub-queries, and then aggregates information into reports. We experiment with one query (i.e., no subquery generation) and three queries (i.e., generating two subqueries) with this pipeline.

Bullet List [64] is an extractive system that first generates 10 Google-like queries, retrieves the top 5 documents per query, extracts key related facts from each document, and finally groups the facts with an LLM. The pipeline is implemented with DSPy [28].

Finally, we implement an iterative system with LangGraph [17] that iteratively generates sub-queries, retrieves, and compresses documents. It uses reflection to identify knowledge gaps and trigger additional retrieval loops, then drafts and revises responses. Since this pipeline is computationally expensive, we only experiment with NeuCLIR24.

For WikiVideo, we use CAG [38] with a Qwen2.5-VL-72B backbone [5]. CAG takes a query and the top 10 videos from the retrieval stack as the input, first extracts key video information related to the query from each video, and then aggregates them into a single response, much like Bullet List but without subquery generation.

4.4 Retrieval Evaluation

To study the relationship between retrieval and RAG, we independently evaluate retrieval models outside of the RAG pipelines to assess the system effectiveness. We evaluate on both coverage-based and relevance-based metrics.

To measure information coverage, we derive nugget qrels from the nuggets in each RAG collection. Specifically for RAG24, since the nugget-document alignment (i.e., the mapping of nuggets to documents containing them) was never recorded (nor completely assessed), we use the same Llama 70B model to judge each relevant document for each nugget in the topic. When assuming a document is relevant if it contains a nugget, our LLM-Judge achieves a precision of 69% and a recall of 90% across all 55 topics compared to the RAG24 relevance judgments. Given that the relevance assessment in RAG24 does not require a document to be considered relevant based on nuggets residing in it, we believe this accuracy is reasonable, and the LLM-judged nugget-document alignment can be used to evaluate retrieval effectiveness for the purpose of this study.¹

We report three coverage metrics, α -nDCG, nDCG using the nugget-based qrels, and Subtopic Recall (StRecall). We use a rank cutoff at 20 for NeuCLIR24 and RAG24, whereas we use a cutoff of 10 for WikiVideo since there are at most 10 relevant videos for each query by design.

The key differences between these three metrics are the penalty for retrieving redundant nuggets (i.e., nuggets that were already present in another document ranked higher) and how redundancy and document positions affect the score. α -nDCG discounts the gain by a document’s rank and reduces the gain when a nugget has already been covered by an earlier document. nDCG uses nugget-based qrels without any consideration of whether a nugget was already covered, which means that only documents that contain a nugget contribute to the score, and there is no penalty for nugget redundancy. StRecall is a set measure without any penalty on the ranking that measures the fraction of nuggets covered.

Additionally, we report relevance metrics using nDCG with the same cutoff on all datasets. We use the qrels that contain the relevance assessments by TREC assessors for relevance-based evaluation on NeuCLIR24 and RAG24, which means that documents that are labeled relevant do not necessarily contain a nugget. In the tables, we use (*R*) to indicate the nDCG metric reporting is using the relevance-based qrels instead of nugget-based when label sources are not explicitly mentioned for space reasons.

4.5 RAG Evaluation

For NeuCLIR24 and RAG24, we use Auto-ARGUE [57] as the primary evaluator for the RAG responses. It takes nuggets in the form of question-answer pairs, which natively come with NeuCLIR24

in the dataset. For RAG24, where the nuggets are in the form of claims, we again used Llama-70B-Instruct to separate them into questions and answers.² With human verification, the nuggets are reasonable and faithfully represent the information in the original claims. We evaluate generated responses using Nugget Coverage, which is the proportion of the grounded nuggets covered in the generated response. A nugget is grounded if it is accompanied by a citation to a document containing the nugget.

For WikiVideo, we use MiRAGE [39], which is its official evaluator. MiRAGE is a nugget-based evaluator for multimodal RAG, measuring factuality, information coverage, and citation support. We report Information Precision (InfoP), evaluating factuality, and Information Recall (InfoR), evaluating coverage.

Note that the definition of InfoR in MiRAGE and Nugget Coverage in Auto-ARGUE is slightly different. InfoR directly assesses whether the nugget is covered in the response, where Nugget Coverage rewards only nuggets in the response with appropriate citation. This slight difference introduces some nuanced analysis, which we will discuss in the next section.

To measure the relationship between metrics, we use Pearson correlation coefficient between metrics. Since we would like to understand whether improvement in retrieval effectiveness is an indicator of improvement in the final RAG responses, we use Pearson correlation instead of rank correlation to directly capture the value relationship instead of system ranking, which is subject to the choice of and the differences between the systems.

5 Analysis

Tables 1 and 2 summarize the effectiveness of models included in this study on both the TREC NeuCLIR 2024 Report Generation Pilot Task (NeuCLIR24) and TREC RAG 2024 (RAG24). The 15 retrieval stacks cover a wide spectrum of retrieval quality, measured by four different metrics, from BM25 generally being the least effective in all metrics to the Qwen3-8B and Rank1-7B reranker being the most effective on NeuCLIR24 and RAG24, respectively. These retrieval models provide the basis for our studies. Similarly, the four RAG pipelines using all 15 retrieval stacks also exhibit varying coverage on nuggets. While the absolute value range is smaller than retrieval, these systems are representative of the common RAG pipelines in the literature, where each represents a particular style of design with its own advantages. Here, Bullet List generally provides the highest nugget coverage for a given retrieval stack in both NeuCLIR24 and RAG24, while GPT-R with one and three queries varies in its ability to cover nuggets in the generated response with different retrieval stacks. This clearly demonstrates the noisy nature of evaluating RAG pipelines when the focus is informational instead of the ability of the underlying language model. Nevertheless, these two tables provide a summary of the underlying data that we use for further relationship analysis in this work.

5.1 Topic-Level Analysis

With all retrieval and generation combinations, we first calculate the topic-level correlation on each benchmark. Table 3 summarizes the Pearson correlation coefficient of the nugget coverage and the

¹Since RAG24 did not publish an overview paper, we instead directly communicated with TREC organizers to verify. We will release the prompt and the judged labels upon publication.

²Prompts and the resulting question-answer nuggets will be released upon publication.

Table 1: Retrieval Evaluation on NeuCLIR24 and RAG24. Columns marked with (R) indicate that the qrels for that metric are the relevance-based qrels instead of the nugget-based ones. All metrics in this table use a rank cutoff of 20.

First Stage	Reranker	NeuCLIR24 Report Generation Pilot Nuggets				RAG24 Nuggets			
		α -nDCG	nDCG	StRecall	(R) nDCG	α -nDCG	nDCG	StRecall	(R) nDCG
BM25	–	0.349	0.328	0.545	0.170	0.450	0.207	0.708	0.263
	Qwen3-8B	0.583	0.581	0.684	0.346	0.633	0.318	0.819	0.410
	Rank1-7B	0.510	0.520	0.629	0.293	0.646	0.343	0.804	0.369
PLAID-X	–	0.446	0.612	0.637	0.295	0.709	0.364	0.919	0.493
	Qwen3-8B	0.634	0.788	0.784	0.416	0.718	0.385	0.903	0.606
	Rank1-7B	0.586	0.762	0.763	0.411	0.730	0.413	0.904	0.542
LSR	–	0.658	0.766	0.853	0.351	0.667	0.317	0.874	0.473
	Qwen3-8B	0.720	0.892	0.874	0.496	0.697	0.365	0.883	0.578
	Rank1-7B	0.651	0.877	0.848	0.458	0.733	0.412	0.908	0.537
Qwen-8B Embed	–	0.627	0.819	0.836	0.390	0.683	0.401	0.899	0.547
	Qwen3-8B	0.691	0.860	0.839	0.468	0.705	0.394	0.903	0.605
	Rank1-7B	0.613	0.820	0.786	0.387	0.742	0.454	0.935	0.548
3 Way RRF	–	0.637	0.796	0.851	0.374	0.705	0.366	0.899	0.587
	Qwen3-8B	0.702	0.870	0.852	0.482	0.709	0.382	0.906	0.623
	Rank1-7B	0.653	0.870	0.840	0.430	0.739	0.432	0.896	0.609

Table 2: Nugget Coverage of RAG responses on NeuCLIR24 and RAG24 using AutoARGUE.

First Stage	Reranker	NeuCLIR24 Report Generation Pilot				RAG24 RAG Task		
		GPT-R (1)	GPT-R (3)	Bullet List	LangGraph	GPT-R (1)	GPT-R (3)	Bullet List
BM25	–	0.449	0.527	0.570	0.490	0.496	0.512	0.602
	Qwen3-8B	0.541	0.509	0.602	0.559	0.510	0.263	0.618
	Rank1-7B	0.484	0.561	0.601	0.574	0.512	0.548	0.559
PLAID-X	–	0.504	0.523	0.586	0.545	0.599	0.588	0.647
	Qwen3-8B	0.583	0.562	0.607	0.545	0.585	0.597	0.652
	Rank1-7B	0.588	0.588	0.581	0.552	0.589	0.551	0.623
LSR	–	0.518	0.568	0.639	0.552	0.568	0.589	0.654
	Qwen3-8B	0.568	0.547	0.599	0.569	0.519	0.571	0.664
	Rank1-7B	0.579	0.599	0.584	0.534	0.587	0.586	0.635
Qwen-8B Embed	–	0.587	0.582	0.608	0.560	0.592	0.603	0.662
	Qwen3-8B	0.571	0.546	0.583	0.532	0.562	0.579	0.658
	Rank1-7B	0.587	0.585	0.604	0.520	0.557	0.594	0.652
3 Way RRF	–	0.600	0.575	0.606	0.547	0.608	0.611	0.662
	Qwen3-8B	0.611	0.599	0.597	0.546	0.582	0.620	0.662
	Rank1-7B	0.582	0.550	0.610	0.490	0.570	0.579	0.649

retrieval effectiveness using the respective retrieval metric. Regardless of the generation pipeline, all generated responses correlate with α -nDCG using nugget-based labels, the highest (with the exception of Bullet List on NeuCLIR24, where correlation is equally high across all nugget-based metrics), indicating that RAG pipeline integrating a retrieval system that is capable of retrieving a wide coverage of information is likely to produce a response that also has a high coverage in information. Nugget-based nDCG@20 also provides a strong indicator. However, since it does not penalize the redundancy of retrieving documents with duplicated information, it

may overestimate the usefulness of a document for the downstream generation model in a ranking, thus leading to lower correlation to the nugget coverage. Interestingly, StRecall, which is a set-based metric, in general, correlates more strongly with the nugget coverage than nDCG@20. While it also does not explicitly penalize the model for including redundant documents, it directly evaluates the set of documents, similar to how the downstream generation model uses the retrieval results. Therefore, this is a metric that aligns with how retrieval results are being used in the pipeline. Since each generation pipeline consumes the retrieval results differently,

but always takes documents from the top of the ranking, α -nDCG provides the most robust evaluation across downstream pipelines.

These three metrics all evaluate the retrieval model on the same information objective, which is coverage. We can also compare the correlation of RAG nugget coverage with the relevance-based metric that evaluates the retrieval stacks. The last row in Table 3 presents such correlations. Note that although the topics are still paired between retrieval and generation evaluation, the objectives are now different, where the labels used in retrieval evaluation are the relevance of each document.³ Therefore, this correlation essentially asks the question: Does a retrieval model that retrieves more relevant documents to the topic provide useful information for downstream generation pipelines to improve nugget coverage? In NeuCLIR24, where information needs are complex and rich report requests, higher relevance does not indicate higher nugget coverage (low correlations). While in RAG24, where information needs are short queries, relevance-based nDCG is still a reasonable indicator for the final nugget coverage, likely because of the kind of information needs are more likely to be satisfied by a single, very relevant document, thus high coverage is not always needed from the retrieval stack.

Overall, to directly answer our RQ1: *For a given RAG pipeline, does an input rank list that covers more information lead to a more effective generated response for a given topic?* Yes, with the strong correlation between nugget-based metrics on the retrieval ranked lists and the nugget coverage on the downstream generated responses, we conclude that input ranked lists with higher information coverage tends to lead to generated responses with higher nugget coverage.

5.2 System-level Analysis

While topic-level analysis is useful, it is not practical when designing a RAG pipeline. Practitioners often select components based on external benchmark evaluations on queries different from those issued to the actual application. Table 4 presents the correlation on the *system* level, where each metric is first averaged over the topics in the respective benchmark before calculating the correlation.

For the two in-task sets of correlations (yellow blocks), nugget-based retrieval metrics provide even stronger correlations, indicating that a retrieval system that is generally effective in retrieving wide coverage of information for a specific task would be a good candidate as the upstream retrieval model in a RAG pipeline. Compared to topic-level correlations, the relationship shown in Table 4 treats the retrieval metric as a property of the retrieval stack, which is untied from any specific topic. We also observe a similar relationship to the relevance-based retrieval metrics, while relevance-based nDCG@20 on RAG24 shows slightly stronger correlation to the nugget coverage than α -nDCG.

System-level analysis allows us to investigate this relationship across different collections and even different tasks (uncolored blocks in Table 4). When evaluating retrieval on RAG24 RAG task and comparing nugget coverage on NeuCLIR24 (lower left block), it shows lower but still strong correlations compared to evaluating

with the same collections. If we move even further away to RAG24 Retrieval task (the last row in Table 4), the correlation becomes lower except for GPT-R with one query. This exception is likely due to the nature of GPT-R when using only one query. It ingests the top 50 documents from the single retrieval result, so nDCG at that point essentially evaluates how well the model recalls all the relevant documents, resulting in a very similar correlation as the nugget-based nDCG. We observe very similar trends on the RAG24 RAG task (the right portion of Table 4).

These results indicate that for the two conditions that we experimented with (retrieval objectives and evaluation), matching both on retrieval and RAG evaluation leads to the strongest correlation. Relaxing one gives a similar level of correlation, where the detailed values vary across RAG pipelines. When mismatching both, retrieval evaluation becomes less useful as an early indicator of the final RAG nugget coverage, especially for iterative pipelines such as LangGraph, with essentially no correlation ($r=0.0941$). Since the RAG queries in RAG24 is less complex, evaluating retrieval effectiveness from any aspect could provide some level of indication.

Therefore, to answer RQ2: *For a given RAG pipeline, does using a more effective retrieval system as a component lead to a more effective RAG system?* Yes, particularly when measuring retrieval systems with a metric that matches the objectives of the final RAG systems. Evaluating on different benchmarks or with different objectives, is still information, though with lower correlations.

Looking across RAG pipelines, similar to the topic-level analysis, GPT-R with one query shows a strong correlation to the nugget-based retrieval metrics, while using three queries (two generated based on the input queries) starts to deviate. Furthermore, Bullet List generates 10 queries from the original user input to try covering various aspects the user may want to see, which exhibit even lower correlation across all retrieval metrics compared to GPT-R on NeuCLIR24. In RAG24, again, since the queries are shorter and factual, Bullet List essentially performs query expansion with parametric knowledge in the underlying LLM. The net effect becomes query expansion instead of improving information coverage since there are only a few aspects to cover. LangGraph has an even more complex interaction with the retrieval model, where it iteratively ingests documents and generates queries to fill in the missing information, leading to lower correlations to the retrieval metrics. While LangGraph may not be the most effective RAG pipeline among the pipelines we have included based on Table 2, it is the most detached one from the retrieval effectiveness. It is capable of achieving nugget coverage of 0.559 with Qwen3-8B Reranker reranking BM25, which is among the highest across its 15 variants using different retrieval stacks, but only 0.532 when using Qwen3-8B Embed followed by Qwen3-8B Reranker. However, BM25 followed by Qwen3-8B Reranker only achieves 0.583 α -nDCG@20, which is much lower than Qwen3-8B Embed followed by Qwen3-8B Reranker (0.691).

This detachment leads to a different design pattern: the LLM needs to adapt to the retrieval model to issue queries that the retrieval model can understand. For example, when using a lexical retrieval model such as BM25, the LLM needs to produce lexical queries for exact matches to the pieces of information that the users need; for retrieval models that are capable of understanding semantic queries, such as Qwen3 Embed, the LLM can form conceptual queries to get fuzzy matches from the retrieval. We observe

³NeuCLIR24 explicitly tasked the annotators to assess how useful a particular document is for writing a report without considering each piece of information in the document, while RAG24 simply asked the annotators to assess the relevancy.

Table 3: Topic-level Pearson correlation coefficients between RAG Nugget Coverage and respective retrieval metrics. Each value for NeuCLIR24 Pilot and RAG24 is calculated based on 285 (19 topics $\times$ 15 retrieval stacks) and 880 (55 topics $\times$ 15 retrieval stacks) pairs of values, respectively. All metrics in this table use a rank cutoff of 20.

Label	Metric	NeuCLIR24 Pilot				RAG24		
		GPT-R (1)	GPT-R (3)	Bullet List	LangGraph	GPT-R (1)	GPT-R (3)	Bullet List
Nugget	α -nDCG	0.5586	0.3489	0.2645	0.3343	0.4419	0.3785	0.3153
Nugget	nDCG	0.4329	0.2714	0.2623	0.1629	0.3114	0.2564	0.1857
Nugget	StRecall	0.4946	0.2907	0.2694	0.2216	0.3805	0.3231	0.2844
Relevance	nDCG	0.1407	-0.0131	0.0458	-0.0239	0.3467	0.3090	0.2881

Table 4: System-level Pearson correlation between RAG Nugget Coverage and respective retrieval metrics and tasks. Cells with a yellow background indicate correlations on the same task (i.e., same objective) and same benchmark; ones with a purple background indicate correlations on the same benchmark but different tasks (i.e., different objectives). All correlations are calculated across RAG pipelines using 15 different retrieval stacks. All metrics in this table use a rank cutoff of 20.

Retrieval Task	Metric	NeuCLIR24 Pilot				RAG24		
		GPT-R (1)	GPT-R (3)	Bullet List	LangGraph	GPT-R (1)	GPT-R (3)	Bullet List
NeuCLIR24 Pilot Report Generation	α -nDCG	0.8105	0.4894	0.4625	0.2893	0.3543	0.2652	0.7580
	nDCG	0.8810	0.5900	0.3346	0.1360	0.5918	0.4855	0.9391
	StRecall	0.8251	0.5886	0.4642	0.1608	0.5272	0.4502	0.8459
NeuCLIR24 MLIR	nDCG	0.8518	0.4960	0.1721	0.2349	0.4061	0.3037	0.8160
RAG24 RAG Task	α -nDCG	0.7792	0.5048	0.2963	0.2557	0.6859	0.4027	0.8745
	nDCG	0.7953	0.5629	0.1572	0.0726	0.6053	0.4116	0.8391
	StRecall	0.7890	0.5161	0.2709	0.1941	0.7923	0.5060	0.8933
RAG24 Retrieval	nDCG	0.9028	0.4915	0.2390	0.0941	0.6938	0.5149	0.9130

Table 5: System-level Pearson correlation between InfoR from MiRAGE and retrieval metrics on NeuCLIR24. The top row additionally reports the macro-averaged InfoR values across all 15 retrieval stacks for each RAG pipeline. Retrieval metrics reported in this table use a rank cutoff of 20.

	GPT-R (1)	GPT-R (3)	Bullet List	LangGraph
Avg. InfoR	0.7279	0.7387	0.6952	0.7325
NeuCLIR24 Pilot				
α -nDCG	0.6777	0.3055	0.4623	0.2050
nDCG	0.6581	0.2546	0.6290	0.1069
StRecall	0.5874	0.3626	0.6050	0.2291
(R) nDCG	0.6869	0.2249	0.4227	0.1434
RAG24				
α -nDCG	0.6626	0.1243	0.6121	-0.0301
nDCG	0.7450	0.1829	0.6047	-0.2034
StRecall	0.5420	-0.0160	0.6484	-0.0930
(R) nDCG	0.7100	0.1395	0.5226	0.0364

this behavior in our LangGraph self-reflection process. The focus of development in improving the final generation output moves away from improving the underlying retrieval model, but rather the adaptivity of the LLM to the specific retrieval model. This may be preferable for some applications or production environments where

the collection is large or only a limited set of retrieval systems is available. Based on our experiments, a simpler (or less iterative) RAG pipeline benefits from the improvement of the retrieval model more directly, which may be more preferable for most applications since finetuning LLM for a specific task or use case is more challenging and computationally expensive than adapting a more effective retrieval model, which is usually more economical.

So to answer RQ3: *Can a more complex RAG pipeline compensate a less effective retrieval system?* Potentially, but it is not guaranteed. Using a more complex RAG pipeline can detach the final response quality from the effectiveness of the underlying retrieval model, as we have shown with our correlation analysis. However, detachment does not always lead to improvements in the final generation quality, as shown in Table 2. The key performance bottleneck in this case shifts from retrieving useful documents to better interaction from the LLM to the retrieval model.

5.3 Using a Different RAG Evaluator

To further understand the robustness of this relationship, we employ MiRAGE as an alternative RAG evaluator to evaluate the generated responses on the NeuCLIR24 task. For brevity, we only present the system-level Pearson correlation between Information Recall (InfoR) from MiRAGE and the retrieval metrics in Table 5. We omit the detailed InfoR scores from the paper and only present the averaged InfoR scores over the 15 retrieval stacks for each RAG pipeline. The overall trend is similar except that MiRAGE prefers LangGraph slightly more than Auto-ARGUE.

Despite slightly different preferences and slightly different metric

Table 6: Multimodal Retrieval and Report Generation Evaluation Results on WikiVideo using MiRAGE. All retrieval metrics use a rank cutoff at 10 since there are at most 10 relevant videos for each topic.

First Stage	Reranker	MultiVENT 2.0		WikiVideo Retrieval				WikiVideo Gen	
		Recall	nDCG	α -nDCG	nDCG	StRecall	(R) nDCG	InfoP	InfoR
CLIP	–	0.333	0.306	0.538	0.498	0.724	0.498	81.5	89.0
	ReasonRank	0.477	0.478	0.628	0.629	0.826	0.629	91.9	87.6
Language Bind	–	0.355	0.326	0.476	0.457	0.634	0.457	84.0	89.2
	ReasonRank	0.498	0.487	0.566	0.563	0.754	0.563	94.5	86.6
Video-ColBERT	–	0.341	0.422	0.431	0.383	0.634	0.383	83.8	89.9
	ReasonRank	0.448	0.535	0.553	0.522	0.734	0.522	87.1	87.6
OmniEmbed	–	0.523	0.495	0.530	0.454	0.721	0.454	88.0	90.9
	ReasonRank	0.590	0.566	0.584	0.587	0.778	0.587	91.2	88.6
MMMORRF	–	0.611	0.585	0.540	0.503	0.724	0.503	94.4	88.0
	ReasonRank	0.634	0.639	0.605	0.617	0.785	0.617	91.8	88.2

Table 7: System-level Pearson correlation between retrieval metrics and Info metrics on WikiVideo. Retrieval metrics reported in this table use a rank cutoff of 10.

Retrieval Task	Metric	InfoP	InfoR
WikiVideo Retrieval	α -nDCG	0.6476	-0.5821
	nDCG	0.6528	-0.6825
	StRecall	0.6530	-0.5405
	(R) nDCG	0.6528	-0.6825
MultiVent 2.0	Recall	0.8447	-0.2837
	nDCG	0.7799	-0.3269

definitions, correlations between InfoR and nugget-based retrieval metrics are still very strong. When comparing with relevance-based nDCG on NeuCLIR, the correlations are still high across RAG pipelines. Interestingly, when evaluating retrieval with RAG24 tasks, we still observe the strong correlation on GPT-R with one query, but the correlations on LangGraph drop from slightly positive to zero with AutoARGUE down to zero to slightly negative. This indicates that under a more lenient RAG information coverage metric (i.e., a metric that does not ignore the information when a sentence is not supported by its citation), LangGraph information is even more detached from the retrieval model, while other systems still exhibit positive correlation with the retrieval metrics.

Therefore, to answer RQ4: *Do these relationships hold across different RAG evaluators?* Yes, the relationship between retrieval and RAG information coverage evaluation still holds when using a different evaluator. However, the nuanced difference between evaluators and their metrics may lead to slightly different results since they are measuring the systems from different aspects and based on slightly different criteria.

5.4 Multimodal RAG

To further validate our findings, in Table 6, we summarize the effectiveness of the multimodal retrieval and generation systems on the WikiVideo video retrieval and article generation task [38]. The 10

retrieval combinations offer a variety of retrieval quality and variance in both human- and claim-based relevance evaluation. Because of the computational cost of multimodal RAG systems, we study only one representative video RAG system using all 10 retrieval stacks. In Table 6, the 10 retrieval stacks exhibit a wide range of retrieval effectiveness on both an external collection (MultiVent 2.0) and the same collection (WikiVideo) as the generation task.

In Table 7, we perform a system-level analysis on the Pearson correlation coefficient of MiRAGE’s Information Precision (InfoP, measuring factuality) and Information Recall (InfoR, measuring coverage) and retrieval effectiveness. Overall, we observe a strong correlation between the factuality of a generation system and retrieval performance instead of information coverage. As shown in Martin et al. [38], multimodal language models are often overfitted on parametric knowledge, resulting in a high tendency in leveraging the parametric knowledge for responding to user queries instead of investigating retrieved materials, similar to the effect of LangGraph in the text RAG analysis. Since topics in WikiVideo are prominent events from 2015 to 2023, information on these events is almost certainly part of the pretraining data. The purpose of retrieval, in this case, becomes verifying the parametric knowledge for improving the factuality of the generation instead of coverage. Such intuition aligns with our observation in Table 7. Nevertheless, retrieval effectiveness is still a strong indicator for the quality of the generation responses.

Overall, to directly answer our RQ5: **Do these relationships hold in multimodal RAG?** Yes, with the strong positive correlation between factuality and retrieval effectiveness. We expect that when evaluating multimodal RAG benchmarks that actually require gathering information from the collection (which, to our knowledge, does not exist yet), information coverage would again correlate with the retrieval information coverage.

6 Conclusion

We investigate the relationships between upstream retrieval effectiveness and downstream RAG information coverage across multiple retrieval systems, RAG pipelines, and evaluation frameworks. We show that nugget-based retrieval metrics correlate strongly with

nugget coverage on the downstream generated response on both topic and system levels, indicating that employing more effective retrieval systems leads to better generated responses, particularly when the retrieval and generation objectives are aligned. More complex or iterative RAG pipelines can decouple this relationship. These results enable more efficient RAG system development by establishing retrieval metrics as reliable proxies for generation quality, reducing the need for costly end-to-end evaluation.

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